

1 · Truth tables — all six gates (2-input)							
A	B	AND A·B	OR A+B	NAND (A·B)′	NOR (A+B)′	XOR A⊕B	XNOR A⊙B
0	0	0	0	1	1	0	1
0	1	0	1	1	0	1	0
1	0	0	1	1	0	1	0
1	1	1	1	0	0	0	1

Fast recall: AND → 1 only at (1,1) · OR → 0 only at (0,0) · NAND = AND inverted · NOR = OR inverted
XOR → 1 in the two middle rows (inputs **differ**; odd count of 1s) · XNOR → 1 at (0,0) & (1,1) (inputs **same**)

2 · Properties matrix — which law does each gate obey?						
Property	AND	OR	NAND	NOR	XOR	XNOR
Idempotent X op X = X	YES X·X=X	YES X+X=X	NO $X\uparrow X=X'$	NO $X\downarrow X=X'$	NO $X\oplus X=0$	NO $X\odot X=1$
Commutative A op B = B op A	YES	YES	YES	YES	YES	YES
Associative (A op B) op C	YES	YES	NO	NO	YES	YES ★trap
Distributive	YES over OR & XOR	YES over AND	NO	NO	NO over AND *	NO
Identity element e X op e = X	e = 1	e = 0	none	none	e = 0	e = 1
Constant case (output independent)	$X\cdot 0 = 0$	$X+1 = 1$	$X\uparrow 0 = 1$	$X\downarrow 1 = 0$	—	—
Complement law X op X′	$X\cdot X'=0$	$X+X'=1$	$X\uparrow X'=1$	$X\downarrow X'=0$	$X\oplus X'=1$	$X\odot X'=0$
Universal gate?	NO	NO	YES	YES	NO	NO

* BUT: AND distributes over XOR → $A\cdot(B\oplus C) = A\cdot B \oplus A\cdot C$ · Summary: all commutative · associative = AND, OR, XOR, XNOR (A-O-X-X) · idempotent = only AND, OR · universal = only NAND, NOR · XNOR is associative (trap!)

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De Morgan · universal recipes · XOR/XNOR magic · traps

De Morgan — NAND and NOR duals

$(A \cdot B)' = A' + B'$

$(A + B)' = A' \cdot B'$

NAND = OR of inverted inputs · NOR = AND of inverted inputs

Universal recipes

NOT = $A \uparrow A = A \downarrow A$

OR from NAND: $(A \uparrow A) \uparrow (B \uparrow B)$

AND from NAND: $(A \uparrow B) \uparrow (A \uparrow B)$

AND from NOR: $(A \downarrow A) \downarrow (B \downarrow B)$

OR from NOR: $(A \downarrow B) \downarrow (A \downarrow B)$

XOR magic (parity gate)

$X \oplus 0 = X \cdot X \oplus 1 = X'$ (programmable inverter)

$X \oplus X = 0 \cdot X \oplus X' = 1 \cdot A \oplus B \oplus B = A$

n-chain XOR = 1 iff odd number of 1s

XNOR (equality detector)

1 iff inputs equal · complement of XOR

n-chain XNOR = 1 iff even number of 1s

Associative (unlike NAND/NOR!)

★ Traps that appear in MCQs

- $X \oplus X = 0$ — XOR is NOT idempotent (same for XNOR: $X \odot X = 1$)
- NAND/NOR commute, but are NOT associative and NOT distributive
- **XNOR IS associative** — $(A \odot B) \odot C = A \odot B \odot C$
- Distributive both ways: only AND and OR
- Universal: NAND alone, NOR alone — XOR is NOT universal

40-second recall script — "Six basic gates: AND (1 iff all 1s), OR (1 iff any 1), NAND/NOR their inverses — both universal. XOR = parity, 1 when inputs differ; XNOR = equality. Properties: all commutative; associative = AND, OR, XOR, XNOR — NAND/NOR are not; idempotent = only AND, OR since $X \text{ XOR } X = 0$. De Morgan: $(AB)' = A' + B'$. Example: full adder Sum = $A \oplus B \oplus C_{in}$, Carry = $AB + C_{in}(A \oplus B)$."